# Macroeconomics I — MPE Insper

Practice list: GDP measurement — three approaches that must agree (Prova 01, Question 1)

**Instructions.** Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Parameters are symbols with the stated admissible ranges; answers are closed forms, conditions, signs and limits.

**How to use this list.** Book exercises are reproduced verbatim, in the book's own notation and currency, after a source line; parts this list adds are marked. Every numerical item asks for GDP by the *product* (value-added), *income* and *expenditure* approaches; an answer that gives one approach only is incomplete, and the three totals must agree. Lay out each answer as Kurlat does: one table with columns Production | Income | Expenditure. Notation: $Y = C + I + G + X - M$ (Kurlat 2020, eq. 1.1.1, p. 15); A = input firm, B = final-goods firm, N = ad-financed social network, as in Prova 01. Amounts without a unit are in reais.

## Block 1 — Exam level: the three approaches, item by item

*(Kurlat §1.1; Jones ch. 2 • 30 points)*

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**Question 1.** (20 points) *Source: Kurlat, P. (2020),* A Course in Modern Macroeconomics*, Exercise 1.1 "Accounting", p. 27 — statement reproduced verbatim, all items (a)–(g).*

**Context from the book (added by this list, from Kurlat §1.1, pp. 15–20):** GDP records the value of everything produced in the country in a year, and can be measured in three equivalent ways (p. 15): *production* (by sector: agriculture & mining, construction, manufacturing, services, government), *income* (employee compensation, proprietor's income — the income of the self-employed —, rental income, corporate profits, interest income, depreciation) and *expenditure*, written as

$$Y = C + I + G + X - M, \tag{1.1.1}$$

where $Y$ is GDP, $C$ consumption, $I$ investment, $G$ public goods and services, $X$ exports and $M$ imports (p. 15). Conventions the items rely on: an input used up in production is not counted again — each producer adds only its *value added*, the value of its output minus the inputs it used up (p. 17); an increase in *inventories* (finished goods, inputs or half-finished products held to be used later) is a form of investment, and when they are sold and go back to zero this is recorded as negative investment (pp. 17–18); GDP includes everything produced within the country, whoever uses it, and excludes goods produced abroad even if they are used in the country (p. 19); government services with no real price are valued at what it cost to produce them (p. 20); a government payment that buys no goods or services (e.g. a pension) is a *transfer*, with no impact on any of the accounts (p. 20).

*What this trains:* building the Production | Income | Expenditure table for imports, inventories, government output valued at cost, a retail margin, investment goods and taxes — the exact skill Prova 01 Q1(b)–(c) asked for.

How does GDP accounting record the following events? For each of them, describe how they would be computed in GDP accounts using the income method, the production method and the expenditure method.

(a) A car manufacturer buys components from Japan for \$1 to be used in production later on and stores them at its warehouse.

(b) A car manufacturer buys components from Japan for \$1 and uses half of those components in the production of a car, that it sells to Andy for \$2. It stores the rest of the components.

(c) An army battalion is deployed to the border to repel a threatened Canadian invasion. The soldiers earn wages of \$10,000 and use ammunition that the government buys for \$5,000. The ammunition is produced using \$2,000 of imported steel and 100 hours of work, for which the workers were paid \$1,000.

(d) Walmart sells 1000 bottles of Coca-Cola for \$1,500. It had previously paid \$1,200 for them.

(e) A shipyard builds a cruise ship. It pays wages of \$200,000, interest on loans (from US residents) of \$100,000 and \$300,000 for imported raw materials. The ship is sold for \$1,000,000 to a cruise company. In the same year, the cruise company has revenue for \$50,000 from operating cruises,

pays wages of \$20,000 to its workers and has no other expenses. Half the cruise revenue comes from tourists who reside in the United States and half comes from tourists who reside abroad.

(f) The government collects \$1000 in income taxes from Roger.

(g) Roger earns \$4000 for working as a babysitter and pays \$1000 in income taxes.

*Solution:*

**Question 2.** (10 points) *Source: Jones, C. I. (2020),* Macroeconomics*, 5th ed., ch. 2, Exercise 2 "What counts as GDP (I)?", p. 39, and Exercise 3 "What counts as GDP (II)?", pp. 39–40 — statements reproduced verbatim, all items.*

**Context from the book (added by this list, from Jones ch. 2, pp. 19–22 and 26):** GDP is "the market value of the final goods and services produced in an economy over a certain period" (p. 19); the production, expenditure and income measures are defined so that they give identical values (p. 20). The expenditure approach is the national income identity

$$Y = C + I + G + NX, \tag{2.1}$$

with $Y$ GDP in dollars, $C$ consumption, $I$ investment (business structures and equipment, intellectual property products such as software and R&D, and the construction of new homes), $G$ government purchases, and $NX$ net exports = exports − imports (pp. 21–22); this list writes it as $Y = C + I + G + X - M$. Government purchases (public schools, highways, research, defense) count; government *spending* also includes transfer payments (Social Security, unemployment insurance) and interest, which are not new production and are not GDP (p. 22). No double counting: each producer contributes its *value added*, its revenue minus the value of the intermediate products it uses, so only the final sale counts; and only *new* production counts — a good that already existed is not counted again when it is resold (p. 26).

*What this trains:* classifying an event in one line — new production or not, final or intermediate, which component of $C + I + G + X - M$ moves — before writing any table.

**2. What counts as GDP (I)?** By how much does GDP rise in each of the following scenarios? Explain.

(a) You spend \$5,000 on college tuition this semester.

(b) You buy a used car from a friend for \$2,500.

(c) The government spends \$100 million to build a dam.

(d) Foreign graduate students work as teaching assistants at the local university and earn \$5,000 each.

**3. What counts as GDP (II)?** By how much does GDP rise in each of the following scenarios? Explain.

(a) A computer company buys parts from a local distributor for \$1 million, assembles the parts, and sells the resulting computers for \$2 million.

(b) A real estate agent sells a house for \$200,000 that the previous owners had bought 10 years earlier for \$100,000. The agent earns a commission of \$6,000.

(c) During a recession, the government raises unemployment benefits by \$100 million.

(d) A new U.S. airline purchases and imports \$50 million worth of airplanes from the European company Airbus.

(e) A new European airline purchases \$50 million worth of airplanes from the American company Boeing.

(f) A store buys \$100,000 of chocolate from Belgium and sells it to consumers in the United States for \$125,000.

*Solution:*

## Block 2 — Harder book items and conceptual classification

*(Kurlat §§1.1, 2.2, 4.3, 5.2, 5.4; Jones ch. 2 • 35 points)*

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**Question 3.** (10 points) *Source: Kurlat, P. (2020),* A Course in Modern Macroeconomics*, Exercise 5.8 "National Accounts and The Golden Rule", pp. 97–98 — statement reproduced verbatim, all items (a)–(d).*

**Context from the book (added by this list, from Kurlat §1.1, p. 21; §§4.1–4.4, pp. 55–67; §§5.1–5.2, pp. 76–77):** *Accounts:* the conventions in the box of Question 1, plus depreciation: machines and buildings lose value over time; GDP is *gross* because it is measured before depreciation, so when a firm subtracts depreciation to compute its profits, the income approach adds it back as a separate line (Ex. 1.10, p. 21). The savings rate is $S/Y = (Y - C)/Y$, savings as a fraction of income, and in a closed economy savings equal investment, $S = I$ (p. 56). *Solow model:* with $Y$ output, $K$ capital, $L$ labor (everyone works, so $L$ is also the population), the Cobb–Douglas production function is $Y = K^{\alpha}L^{1-\alpha}$, $\alpha \in (0,1)$ (4.1.2), p. 55 (writing it $AK^{\alpha}L^{1-\alpha}$, with $A$ the level of technology, "doesn't make much difference", p. 77). The population grows at rate $n$, $L_{t+1} = (1+n)L_t$ (p. 55); people save a constant fraction $s$ of income, so $I = sY$ (4.1.3); every piece of capital loses a fraction $\delta$ of its value each period, so $K_{t+1} = (1-\delta)K_t + I_t$ (4.1.4) (pp. 56–57). Per worker, $y \equiv Y/L$, $k \equiv K/L$ and $y = f(k) \equiv F(K/L, 1)$ (4.2.1) (p. 57). In a steady state ($ss$), $sf(k_{ss}) = (\delta + n)k_{ss}$ (4.3.2) and consumption is $c_{ss} = (1-s)y_{ss}$ (4.3.1); the *Golden Rule* is the saving behavior that maximizes steady-state consumption, and the Golden Rule capital stock $k_{gr}$ satisfies $f'(k_{gr}) = \delta + n$ (4.3.4) (pp. 62–63). *Factor markets* are perfectly competitive, so the rental rate of capital and the wage equal the marginal products, $r^K = F_K(K,L)$ (4.4.8) and $w = F_L(K,L)$ (4.4.9) (p. 66), and all output is paid to workers or to capital owners, $r^K K + wL = F(K,L)$ (Prop. 4.2, p. 67). With Cobb–Douglas the labor share is $wL/Y = 1 - \alpha$ (5.1.1), so the capital share is $\alpha$ (p. 76).

*What this trains:* the three approaches with an investment good and depreciation, then reading factor shares and the saving rate off the accounts and using them in the golden rule — Prova 01 Q1 and Q2(b) in one exercise.

In 2018, the following events happened in Aurum, which is a closed economy that uses the Denarius (plural: Denarii) as its currency.

- Grano, Inc. hired Cornelia to plant and harvest 500 tons of wheat on a large plot of land using a tractor. Both the land and the tractor are owned by Grano, Inc.

  - It paid Cornelia 200 Denarii for her work on the harvest.
  - It sold the wheat to Panem, Inc. for 1,000 Denarii.
- The tractor the Cornelia used:
  - Was new at the beginning of the year. It was worth 2,000 Denarii.
  - Was no longer new at the end of the year. By then it was worth 1,800 Denarii.
- Panem, Inc. hired Gaius to grind the wheat in its mill and produce a million loaves of bread.
  - It paid Gaius 1,400 Denarii for his work.
  - It sold the bread for 4,000 Denarii to the hungry citizens, who eat it.
- Fabrica, Inc. hired Livia to build a combine harvester.
  - It paid Livia 400 Denarii for her work.
  - It sold the combine harvester to Grano, Inc. for 1,000 Denarii.

(a) Construct GDP accounts (in Denarii) for Aurum by production, income and expenditure.

(b) What were the income shares of labor and capital?

(c) What was the savings rate?

(d) Assume the following:

- the savings rate is constant,
- the production function has the form:

$$Y = AK^{\alpha}L^{1-\alpha}$$

  and there is no technological progress,
- the depreciation rate for the tractor is typical for the capital stock as a whole,
- the population is constant.

In the long run, would the level of consumption be higher if the economy slightly increased its saving rate?

*Solution:*

**Question 4.** (12 points) *Source: Kurlat, P. (2020),* A Course in Modern Macroeconomics*, Exercise 5.7 "GDP Accounting, Interest Rates and Growth Accounting", pp. 96–97 — statement reproduced verbatim, all items (a)–(e).*

**Context from the book (added by this list, from Kurlat §4.4, pp. 66 and 69; §5.4, pp. 86–87):** *Accounts:* imports, government output at cost and transfers as in the box of Question 1; depreciation, the savings rate, the Cobb–Douglas function and its factor shares (labor $1-\alpha$, capital $\alpha$) as in the box of Question 3. $F_K$ is the marginal product of capital, $\partial Y/\partial K$. *Interest rate:* in competitive markets the rental rate of capital is $r^K = F_K(K, L)$ (4.4.8) (p. 66); someone who borrows at the real interest rate $r$ to invest in capital earns $r^K - \delta$, so the loan market clears only if

$$r_{t+1} = r^K_{t+1} - \delta \tag{4.4.12}$$

(p. 69), with $\delta$ the rate at which capital loses value. *Growth accounting:* with $Y_t = F(K_t, L_t, A_t)$ (5.4.1), $A$ technology, and GDP growth $g_Y \equiv Y_{t+1}/Y_t - 1$, a first-order Taylor approximation gives (5.4.2): $g_Y \approx$ capital share $\times$ growth rate of capital + labor share $\times$ growth rate of labor + Solow residual, the shares being $F_K K/F$ and $F_L L/F$ (pp. 86–87). The residual is measured as "Solow Residual = GDP growth − Capital share $\times$ Capital Growth − Labor Share $\times$ Labor Growth" and is also called Total Factor Productivity, TFP (p. 87).

*What this trains:* an open economy with imported capital, imported consumption, exports, government output at cost and income taxes; then the same accounts feed the interest rate and a Solow residual.

Proletaria is a mythical country in Central Asia, where the currency is the ruble. The exchange rate is 10 rubles per euro.

Suppose the production function in Proletaria is given by:

$$Y = K^{\alpha} L^{1-\alpha}$$

(a) Find expressions for the output-to-capital ratio $\frac{Y}{K}$ and the marginal product of capital $F_K$ as functions of $K$ and $L$.

During 2014, the following events took place in Proletaria.

- Kapitas, the main manufacturer in the country, imported an industrial welder made in Germany, for which it paid 1,000 euro.
- By the end of the year, the industrial welder was no longer new. Its estimated value in the resale market was 960 euro.
- 100 workers worked for Kapitas the entire year making screws and nails, using the new welder. Each of them received wages for 350 rubles.
- Each of the Kapitas workers paid 100 rubles in income taxes.
- The total output of Kapitas consisted of 1 million screws and 1 million nails. All of it was exported to Austria for a total of 4,000 euro.
- The government of Proletaria employed one of the 100 workers as Chief of the Secret Police (in addition to his factory job) to maintain law and order, and paid her a salary of 10,000 rubles.
- The workers ate beef imported from France, which cost a total of 2,500 euro.

(b) Construct GDP accounts (in rubles) for Proletaria by production, income and expenditure.

(c) What were the income shares of labor and capital?

(d) Suppose that we know that the capital stock in Proletaria is 100,000 rubles and that the depreciation rate of the industrial welder is typical for this country's capital stock. What interest rate should we expect to observe?

(e) During 2015, additional investment in Proletaria has exceeded depreciation so that the capital stock now stands at 120,000 rubles. The labor force also grew thanks to immigration, and now consists of 105 workers instead of 100. GDP during 2015 was 55,000 rubles (prices were

constant). How much of the growth in GDP between 2014 and 2015 can be attributed to growth in Total Factor Productivity?

*Solution:*

**Question 5.** (10 points) Conceptual — answer in words, no numbers. For each event, say whether it adds to this year's GDP; if it does, name the producer whose value added rises, the income it generates and the final use ($C$, $I$, $G$ or $X$) it enters; if it does not, say why. Two to four sentences each.

**a)** A search engine is free for households and is paid for by advertising sold to firms.

**b)** A bank's own IT staff write a risk-management program the bank will use for years; a drug company's laboratory runs a research project for its own future products.

**c)** A household sells its five-year-old car to another household through a used-car dealer.

**d)** One family owns the house it lives in; next door an identical house is rented to a tenant.

**e)** An accounting firm prepares one tax return for a manufacturing company and one for a household.

**f)** The government pays retirement pensions and pays the salaries of public-school teachers.

*Solution:*

**Question 6.** (3 points) *Source: Kurlat, P. (2020),* A Course in Modern Macroeconomics*, Exercise 1.4 "Drugs and Prostitution", p. 29, and Exercise 2.4 "Construction and the Armed Forces", p. 41; Jones, C. I. (2020),* Macroeconomics*, 5th ed., ch. 2, Exercise 9 "Earthquakes and GDP", p. 41 — statements reproduced verbatim; the closing instruction is added by this list.*

**Context from the books (added by this list, from Kurlat §1.1, pp. 15 and 21; Kurlat §2.2, pp. 33–34; Jones ch. 2, pp. 19, 22 and 26):** *Kurlat 1.4.* GDP records the value of everything produced in the country in a year (p. 15); activity that involves no market transaction is usually not included: "Most of the time, however, we compute the value of an activity only if it is sold in the market" (p. 21). The article is not reproduced here; its address names its subject: Italy's inclusion of illicit activities in its GDP figures. *Kurlat 2.4.* Jones and Klenow measure the living standard of a country, Utilia, by the number $\lambda$ that makes a person indifferent between living a year as a random resident of Utilia and as a random resident of the US with everyone's consumption multiplied by $\lambda$ (p. 33), using

$$u(c,l,a)=\mathbb{E}\Big[\Big(\bar{u}+\frac{c^{1-\sigma}}{1-\sigma}-\theta(1-l)^2\Big)a\Big], \tag{2.2.1}$$

$$u\big(\lambda c^{US},l^{US},a^{US}\big)=u\big(c^{Utilia},l^{Utilia},a^{Utilia}\big), \tag{2.2.2}$$

where $c$ is consumption, $l$ leisure (the fraction of time spent on activities not counted in GDP), $a$ equals 1 if alive and 0 if dead, and $\bar{u}$, $\sigma$, $\theta$ are parameters (pp. 33–34). What enters (2.2.1) is consumption, not production: output dedicated to investment or exported gives no utility during the year (investment benefits future residents), and public and private consumption are assumed to give the same utility, so what enters is $C+G$ of (1.1.1) (box of Question 1), more precisely $C$ plus the government-consumption part of $G$ (p. 34). *Jones 9.* GDP is "the market value of the final goods and services produced in an economy over a certain period" (p. 19); the construction of new homes is part of investment (p. 22); GDP counts only new production (p. 26).

*What this trains:* separating what GDP measures (market production) from what it does not (legality, welfare, the stock of wealth).

**Kurlat 1.4 Drugs and Prostitution.** Read the following article: `http://www.economist.com/news/finance-and-economics/21603073-italys-inclusion-illicit-activities-its-figures-excites-much-interest-sex`. What do you think? Should drug production and prostitution be included in the calculation of GDP?

**Kurlat 2.4 Construction and the Armed Forces.** Suppose we compared welfare in two otherwise identical countries. Country A dedicates 50% of GDP to maintaining a large army. Country B dedicates 50% of GDP to building houses. In which of the two countries would welfare be higher, according to the measure used by Jones and Klenow. What do you think of this?

**Jones 9. Earthquakes and GDP:** Suppose the rural part of a country is hit by a major earthquake that destroys 10 percent of the country's housing stock. The government and private sector respond with a major construction effort to help rebuild houses. Discuss how this episode is likely to affect (a) the economic well-being of the people in the country and (b) the economy's measured GDP.

*Added (not in the book):* answer each of the three in words.

*Solution:*

## Block 3 — Created: the Prova 01 economy, extended

*(Kurlat §1.1 • 35 points)*

**Question 7.** (12 points) *Inventories and data sales.* The economy of Prova 01 in a new year. All values in reais; nothing else happens.

- Firm A uses no intermediate inputs. It produces inputs worth 800 and pays wages of 500. It sells inputs worth 600 to B and keeps the remaining 200 in its warehouse.
- Firm B uses all 600 of the inputs it bought, pays wages of 200, buys advertising from N for 100 and a market-research dataset from N for 60. It sells final goods to households for 900; final goods worth 150 at market prices are still unsold in its warehouse at the end of the year.
- Social network N is free for its users. It sells the advertising (100) and the dataset (60) to B and another dataset to a firm abroad for 40. It pays wages of 120 and has no other costs.

**a)** (7 points) Compute GDP by the product, income and expenditure approaches in one table (by firm for product and income, by component for expenditure) and check that they agree.
**b)** (3 points) In words: why do the 200 of stored inputs and the 150 of unsold goods count in this year's GDP, and why does N's data enter the expenditure column once and not the other time?
**c)** (2 points) Next year B sells its 150 of stored goods to households for 150, and nothing else happens. Give the three approaches for that year and explain the result in one sentence.

*Solution:*

**Question 8.** (12 points) *Imports, a software sale, dividends and the government.* Another year of the Prova 01 economy. All values in reais; nothing else happens.

- Firm A imports raw materials for 100, turns them into inputs and sells the inputs to B for 600. It pays wages of 350.
- Firm B uses the inputs, buys advertising from N for 100, pays wages of 250 and sells final goods for 1,000: 850 to domestic households and 150 to buyers abroad.
- Network N is free for its users. It sells the advertising (100) and sells licences of a photo-editing app to domestic households for 80. It pays wages of 150, of which 100 go to a team that builds a recommendation engine N will use itself for several years. It has no other costs. Out of its profit, N pays dividends of 40; half of its shareholders live abroad.
- The government hires content moderators for wages of 30 (their service is free to the public), pays unemployment benefits of 25 and collects taxes of 55 from households.

**a)** (6 points) Compute GDP by the three approaches in one table and check that they agree.
**b)** (2 points) Compute gross national income (GNI) and explain why it differs from GDP.
**c)** (2 points) In words: classify the app licences, the recommendation engine, the dividends and the unemployment benefits, each in one sentence.
**d)** (2 points) Prova 01 Q1(c) had GDP = 1,100. Account for the difference between that number and your answer to (a), item by item.

*Solution:*

**Question 9.** (11 points) *Capitalise or expense? Depreciation and measured growth.* Two years of the Prova 01 economy. A's and B's value added is all paid out as factor income (wages and profits; the split is not needed). All values in reais.

- **Year 1** is Prova 01 Q1(c): A sells inputs to B for 600; B sells final goods to households for 1,000; N sells advertising to B for 100 and uses that 100 to pay workers who build software for N's own use. The software is ready at the end of year 1, will be used for four years and depreciates by $100/4 = 25$ a year from year 2 on.
- **Year 2**: A, B and the advertising are as in year 1, and no new software is built. N launches ad-free premium accounts: households buy accounts for 50, and B buys accounts for 20 for its marketing staff to use at work. N pays its support staff wages of 30 and has no other costs.

**a)** (3 points) Year 1: compute GDP by the three approaches (i) when the software is capitalised (current convention) and (ii) when it is expensed as a current cost of N (the older convention).
**b)** (3 points) Year 2, software capitalised: compute GDP by the three approaches, and net domestic product (NDP).
**c)** (2 points) Compute GDP growth from year 1 to year 2 under each convention, and explain in words why the sign differs.
**d)** (1.5 points) In words: why do the households' 50 of premium accounts enter $C$ while B's 20 do not?
**e)** (1.5 points) In words: a student wrote that the 100 of software is "consumption, because the workers spend their wages". Explain what is wrong, using the three columns.

*Solution:*